

Uncertainty and sentiment in financial reports

FRE-GY 7871 A · NLP and the Investment Process · Filing dates 2021-01-01 to 2025-12-31

Research repository

1,655 filings from 91 companies enter the text analysis. The volatility analysis uses 1,437 filings and the return analysis uses 1,437. Negative words measure adverse language; uncertainty words measure imprecision and hedging. All results describe associations in companies selected from ARK holdings.

Within companies, annual-report negative-word shares change by +0.068 percentage points per year ($p < 0.001$); quarterly-report uncertainty changes by -0.019 points ($p = 0.059$). After controlling for prior volatility, 0 of 6 uncertainty specifications are significant at 5%. The pooled negative-tone return estimate is not significant at 5% ($p = 0.206$).

Filing coverage ends on 2025-12-31; market observations end on 2026-09-08. Recent filings remain in text analysis even when their future price windows are unavailable. Return and volatility samples are filtered separately; both volatility specifications use the same observations. The latest included filings are 2025-12-22 for returns and 2025-12-22 for volatility.

Table 1. Sample construction

Panel A counts holding identifiers, then companies. The frozen six-fund holdings use January 2, 2026 for ARKF and ARKX and September 4, 2026 for the other four funds. Different share classes are combined by SEC company identifier. The holdings dates define a retrospective company universe.

Company/security filter	Removed	Remaining	Unit
Raw holding identifiers	0	130	identifiers
Funds and non-company securities	4	126	identifiers
Foreign local listings	8	118	identifiers
Unresolved SEC identifiers	3	115	identifiers
Combine share classes by company CIK	1	114	companies
No 10-K/Q in selected period	6	108	companies
20-F / 40-F reporting companies	17	91	companies

Panel B — Text sample. Counts follow this panel's filter order.

Filing filter	Removed	Remaining	Companies
All 10-K/Q and amendments	0	1,709	91
Remove amendments and parse failures	46	1,663	91
Minimum words: 2,000 K / 1,000 Q	0	1,663	91
Earliest company filing each quarter	8	1,655	91

Panel B — Volatility sample. Counts follow this panel's filter order.

Filing filter	Removed	Remaining	Companies
Eligible text filings	0	1,655	91
Usable day 0 and prior price at least \$3	108	1,547	91
Outcome window elapsed by market cutoff	0	1,547	91
60 observed returns before and after	23	1,524	90
Complete pre/post volatility windows	0	1,524	90
Accession-matched outstanding shares	70	1,454	85
Complete liquidity and model controls	17	1,437	84

Panel B — Return sample. Counts follow this panel's filter order.

Filing filter	Removed	Remaining	Companies
Eligible text filings	0	1,655	91
Usable day 0 and prior price at least \$3	108	1,547	91
Outcome window elapsed by market cutoff	0	1,547	91
60 observed returns before filing	23	1,524	90
Complete four-session return window	0	1,524	90
Accession-matched outstanding shares	70	1,454	85
Complete liquidity and model controls	17	1,437	84

46 amendments and 0 parse failures are recorded. All 1,663 successfully parsed original filings receive tone scores before analytical sample restrictions. Market-data restrictions do not determine the text sample.

Table 2. Tone by report type

The text sample determines descriptive statistics and trends. Proportional scores are percentages; weighted scores are equation (1) sums. Document frequencies are fitted within each analytical sample and separately by report type.

Form	Measure	N	Mean	SD	P25	Median	P75
10-K	Negative words (%)	406	2.267	0.421	1.990	2.287	2.558
10-K	Uncertainty words (%)	406	1.943	0.239	1.797	1.963	2.116
10-K	Negative tf.idf	406	100.618	36.957	72.126	93.781	127.176
10-K	Uncertainty tf.idf	406	15.887	5.707	11.994	14.996	19.041
10-Q	Negative words (%)	1,249	2.044	0.965	1.168	1.836	2.963
10-Q	Uncertainty words (%)	1,249	1.816	0.586	1.318	1.649	2.408
10-Q	Negative tf.idf	1,249	92.616	78.211	26.592	60.803	155.600
10-Q	Uncertainty tf.idf	1,249	15.931	8.606	9.006	13.969	22.249

Negative/uncertainty correlations (proportional, weighted): All: 0.849, 0.918; 10-K: 0.718, 0.821; 10-Q: 0.855, 0.925.

Table 3. Most frequent dictionary words

Each percentage divides a word count by all token occurrences in its dictionary category across the text sample.

Rank	Negative word	Share %	Uncertainty word	Share %
1	LOSS	5.230	MAY	35.412
2	ADVERSELY	4.201	COULD	18.439
3	LOSSES	3.149	RISK	4.954
4	CLAIMS	3.095	RISKS	4.479
5	ADVERSE	2.823	BELIEVE	2.932
6	AGAINST	2.485	APPROXIMATELY	2.553
7	UNABLE	2.026	ASSUMPTIONS	1.892
8	LITIGATION	1.979	INTANGIBLE	1.609
9	HARM	1.892	POSSIBLE	1.336
10	FAILURE	1.842	UNCERTAINTIES	1.156
11	FAIL	1.365	FLUCTUATIONS	1.107
12	IMPAIRMENT	1.240	MIGHT	1.072
13	DIFFICULT	1.178	UNCERTAIN	1.024
14	NEGATIVELY	1.169	ANTICIPATED	0.975
15	PENALTIES	1.164	PREDICT	0.938
16	NEGATIVE	1.047	VOLATILITY	0.935
17	DELAYS	0.949	DEPEND	0.927
18	RESTATED	0.940	UNCERTAINTY	0.886
19	DECLINE	0.931	DIFFER	0.855
20	DAMAGES	0.926	ANTICIPATE	0.801
21	LIMITATIONS	0.869	CONTINGENT	0.775

Rank	Negative word	Share %	Uncertainty word	Share %
22	DELAY	0.855	EXPOSURE	0.758
23	VOLATILITY	0.789	DEPENDS	0.673
24	CHALLENGES	0.770	VARIABLE	0.667
25	FINES	0.725	PENDING	0.646
26	DISRUPTIONS	0.699	DEPENDENT	0.582
27	HARMED	0.698	CONTINGENCIES	0.548
28	INVESTIGATIONS	0.671	PROBABLE	0.506
29	BREACH	0.668	ASSUMED	0.496
30	INFRINGEMENT	0.623	VARY	0.478

The top ten words account for 28.7% of negative tokens and 74.8% of uncertainty tokens. The active dictionaries contain 2,345 negative and 297 uncertainty words, with 40 words in both categories.

Tone measures

The Loughran–McDonald Master Dictionary, 1993–2025 release, was updated in March 2026. A positive category value identifies an active word; a negative value records removal from that category. Only active entries enter the scores. Thus the negative category contains 2,345 words; the 10 removed entries are excluded. The proportional score divides category occurrences by all tokens in a filing.

$$P_{cj} = \frac{\sum_{i \in c} tf_{ij}}{W_j}$$

Equation (1) applies logarithmic term frequency and inverse document frequency. The score sums weights over category words observed in a filing.

$$T_{cj} = \sum_{i \in c, tf_{ij} > 0} \frac{1 + \ln(tf_{ij})}{1 + \ln(a_j)} \ln\left(\frac{N}{df_i}\right)$$

$$a_j = \frac{W_j}{V_j}$$

W is total tokens, V is distinct tokens, N is documents in the relevant estimation corpus and df is document frequency. All logarithms are natural. The text, volatility and return samples each fit their own word weights; report-type analyses also refit within form. The weighted score is measured in score units rather than percentages.

An illustrative calculation uses “LOSS LOSS RISK GAIN”, “LOSS GAIN GAIN” and “RISK RISK RISK GAIN”, with category {LOSS, RISK}. Their weighted scores are 0.8480, 0.2885 and 0.5026. These constructed documents illustrate the formula.

Primary-document text retains visible XBRL facts and excludes hidden content, separate exhibits and tables with more than 15% digits among non-space characters. Alphabetic tokens contain at least two characters and retain apostrophes and hyphens. Incorporated-by-reference text is not recovered.

TSLA's 10-Q filed 2023-10-23 contains 2.72% negative words and 1.05% uncertainty words, at percentile ranks 72 and 4, respectively. It has the largest negative-minus-uncertainty percentile gap in the text sample. This selected contrast is not representative of all filings.

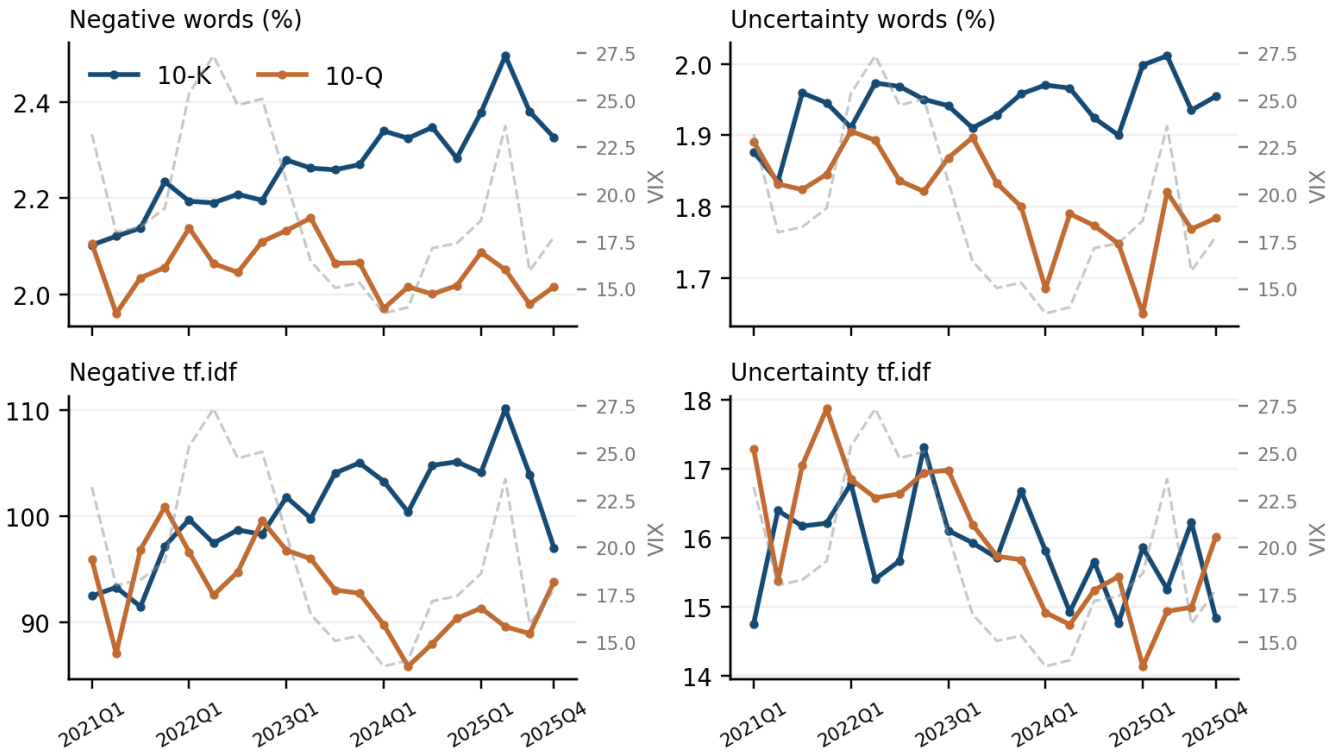
Selected filing language: On October 27, 2021, the Court approved the parties' joint stipulation that, among other things, (a) all claims against Kimbal Musk and Steve Jurvetson ...

Dictionary counts identify category vocabulary; they do not determine the direction or significance of the event described.

MAY and COULD account for 53.9% of uncertainty tokens. These counts do not distinguish recurring disclosure language from newly expressed uncertainty. High negative/uncertainty correlations also mean the two measures do not provide independent evidence.

MAY occurs in 100.0% of filings and APPROXIMATELY in 99.0%. Inverse document frequency gives less weight to words appearing in most documents.

Figure 1. Quarterly tone and VIX



Company-centered means restore each form mean; entry/exit timing can still affect composition.
Dashed gray: mean VIX (right axis). Weights fitted by form.
Minimum 10-K quarter N = 1; quarterly counts accompany the analysis.

10-K: 1–78 filings per observed quarter; 10-Q: 7–90 filings per observed quarter. Sparse annual-report quarters may reflect only a few companies. Company-centered means reduce baseline composition differences, but entry and exit periods can still affect this unbalanced panel. The gray dashed series is quarterly average VIX on the right axes; the latest quarter is partial when indicated. Formal trend inference uses Table 4.

Table 4. Annual tone trends

Quarter-level means estimate the aggregate trend. Filing-level models estimate change within companies:

$$\bar{T}_q = \alpha + \beta \tau_q + \sum_{s=2}^4 \delta_s D_{sq} + \varepsilon_q$$

$$T_{iq} = \alpha_i + \beta \tau_q + \sum_{s=2}^4 \delta_s D_{sq} + \theta K_{iq} + \varepsilon_{iq}$$

Time is elapsed years from the first sample quarter. Seasonal indicators control the reporting quarter of the year. Company effects control company mean levels; K indicates a 10-K in the pooled sample and is omitted within each report type. Saturated calendar-quarter effects are excluded because they would absorb the trend. Aggregate inference shows ordinary OLS and Newey–West t-statistics with four lags. Within-company t and p use two-way company/calendar-quarter clustering.

Sample	Measure	Agg. slope	OLS t	NW t	Within slope	Within t	p
All	Neg. %	0.017	1.733	1.383	0.009	0.763	0.455
All	Unc. %	-0.003	-0.585	-0.572	-0.008	-1.139	0.269
All	Neg. tf.idf	0.994	1.261	1.067	0.093	0.106	0.917
All	Unc. tf.idf	-0.173	-1.221	-0.958	-0.281	-1.931	0.069
10-K	Neg. %	0.056	8.856	9.915	0.068	6.833	1.60×10^{-6}
10-K	Unc. %	0.013	2.140	2.265	0.028	5.751	1.53×10^{-5}
10-K	Neg. tf.idf	2.019	2.419	4.050	2.559	2.979	0.008
10-K	Unc. tf.idf	-0.278	-1.505	-2.605	0.033	0.139	0.891
10-Q	Neg. %	-0.021	-1.561	-1.867	-0.008	-0.557	0.584
10-Q	Unc. %	-0.037	-3.036	-3.418	-0.019	-2.009	0.059
10-Q	Neg. tf.idf	-1.745	-2.342	-2.620	-1.652	-1.401	0.177
10-Q	Unc. tf.idf	-0.545	-4.260	-4.849	-0.480	-3.066	0.006

Proportional slopes are percentage points per year; weighted slopes are score units per year. All: 91 companies, 20 quarters, 19 inference degrees of freedom; 10-K: 88 companies, 20 quarters, 19 inference degrees of freedom; 10-Q: 91 companies, 20 quarters, 19 inference degrees of freedom.

10-K negative-word shares change by +0.068 percentage points per year ($p < 0.001$); uncertainty-word shares change by +0.028 points ($p < 0.001$). The corresponding weighted slopes are +2.559 ($p = 0.008$) and +0.033 ($p = 0.891$).

10-Q negative-word shares change by -0.008 percentage points per year ($p = 0.584$); uncertainty-word shares change by -0.019 points ($p = 0.059$). The corresponding weighted slopes are -1.652 ($p = 0.177$) and -0.480 ($p = 0.006$).

Within-company estimates receive more weight than aggregate slopes because they account for company levels and reporting season. Proportional and weighted measures capture related language and are not independent replications. Sparse annual-report quarters limit interpretation of individual points in the chart.

Filing-event design

Day 0 is the first NYSE session on or after the later of the filing date and the Eastern acceptance date. Acceptance at or after the exchange's actual close, including an early close, moves the event to the next session. Market data include only completed sessions through 2026-09-08.

$$R_{[0, 3], \text{excess}} = \frac{P_{+3}^{\text{adj}}}{P_{-1}^{\text{adj}}} - \frac{SPY_{+3}^{\text{adj}}}{SPY_{-1}^{\text{adj}}}$$

$$\sigma^{\text{pre}} = \sqrt{252 \text{ SD}(r_{-60}, \dots, r_{-6})}$$

$$\sigma^{\text{post}} = \sqrt{252 \text{ SD}(r_{+4}, \dots, r_{+63})}$$

The event spans four daily return intervals. Pre-filing volatility uses 55 daily returns and post-filing volatility uses 60, with sample standard deviations. Adjusted closes measure returns; nominal closes determine the \$3 price filter and company size. Later splits are reversed for nominal price and volume histories. Future returns are not imputed.

Size is the day -1 nominal price times outstanding shares from the scored filing's cover page. Same-date share classes are summed using one class price as a proxy; future or weighted-average shares are excluded. Liquidity is mean nominal dollar volume over days [-60, -6], and prior excess return is SPY-adjusted buy-and-hold return over that interval.

The outcome models use firm effects, calendar-quarter effects and the following controls:

$$C_{iq} = \gamma_1 \ln(\text{Size}_{iq}) + \gamma_2 \ln(\text{DollarVolume}_{iq}) + \gamma_3 R_{iq}^{\text{pre, excess}} + \theta K_{iq}$$

$$\sigma_{iq}^{\text{post}} = \alpha_i + \lambda_q + \beta U_{iq} + C_{iq} + [\rho \sigma_{iq}^{\text{pre}}] + \varepsilon_{iq}$$

$$R_{iq}^{[0, 3], \text{excess}} = \alpha_i + \lambda_q + \beta N_{iq} + C_{iq} + \rho \sigma_{iq}^{\text{pre}} + \varepsilon_{iq}$$

U is uncertainty and N is negative tone, each estimated separately as a proportion and a weighted score. The bracketed pre-volatility term is included or omitted in the paired volatility tests; it is always included in the return tests. The 10-K indicator K applies only to the pooled sample. No winsorisation is applied.

Table 5. Uncertainty and subsequent volatility

Dependent variable: annualised volatility as a fraction. Coefficients are per unit of uncertainty fraction or weighted score. Both specifications use the same complete observations within each sample. Two-way company/quarter clustering determines t and p. All: 84 companies, 20 quarters, 19 inference degrees of freedom; 10-K: 83 companies, 20 quarters, 19 inference degrees of freedom; 10-Q: 83 companies, 20 quarters, 19 inference degrees of freedom.

Sample	Measure	Pre-vol	Coefficient	t	p	N
All	prop	No	-0.169	-0.117	0.908	1,437
All	prop	Yes	-0.721	-0.567	0.578	1,437
All	tfidf	No	1.69×10^{-4}	0.206	0.839	1,437
All	tfidf	Yes	-1.49×10^{-4}	-0.199	0.844	1,437
10-K	prop	No	-16.264	-0.759	0.457	349
10-K	prop	Yes	-24.921	-1.033	0.315	349
10-K	tfidf	No	0.004	0.555	0.586	349
10-K	tfidf	Yes	0.005	0.905	0.377	349
10-Q	prop	No	3.207	1.338	0.197	1,088
10-Q	prop	Yes	2.221	0.975	0.342	1,088
10-Q	tfidf	No	0.003	2.171	0.043	1,088
10-Q	tfidf	Yes	0.002	1.602	0.126	1,088

Proportional uncertainty: a one-standard-deviation increase corresponds to -0.38 percentage points of annualised volatility after controlling for prior volatility (95% interval [-1.77, +1.01]). The uncontrolled effect is -0.09 points ($p = 0.908$), compared with $p = 0.578$ after control. Adding the control changes the estimated effect by -0.29 points. This comparison tests incremental association beyond existing volatility.

Weighted uncertainty: a one-standard-deviation increase corresponds to -0.14 percentage points of annualised volatility after controlling for prior volatility (95% interval [-1.63, +1.34]). The uncontrolled effect is +0.16 points ($p = 0.839$), compared with $p = 0.844$ after control. Adding the control changes the estimated effect by -0.30 points. This comparison tests incremental association beyond existing volatility.

For 10-Q weighted uncertainty, the effect per tone standard deviation changes from +2.21 to +1.60 volatility percentage points after adding prior volatility ($p = 0.043$ and $p = 0.126$). The controlled proportional-score estimate has $p = 0.342$. Differences across weighting schemes and multiple unadjusted tests limit the strength of an isolated significant association.

Table 6. Negative sentiment and filing-period excess return

Dependent variable: SPY-adjusted four-session buy-and-hold return as a fraction. All models include log size, log dollar volume, prior excess return, pre-filing volatility, company effects and calendar-quarter effects; pooled models also include the 10-K indicator. Coefficients are per unit of negative-word fraction or weighted score. t and p use two-way company/quarter clustering. All: 84 companies, 20 quarters, 19 inference degrees of freedom; 10-K: 83 companies, 20 quarters, 19 inference degrees of freedom; 10-Q: 83 companies, 20 quarters, 19 inference degrees of freedom.

Sample	Measure	Coef.	t	p	Effect pp	MDE pp	N
All	prop	-0.958	-1.310	0.206	-0.820	1.837	1,437
All	tfidf	-1.46×10^{-4}	-1.259	0.223	-1.060	2.471	1,437
10-K	prop	-7.560	-0.873	0.394	-3.066	10.308	349
10-K	tfidf	-8.16×10^{-4}	-1.074	0.296	-3.107	8.490	349
10-Q	prop	-2.723	-2.170	0.043	-2.592	3.506	1,088
10-Q	tfidf	-3.53×10^{-4}	-2.106	0.049	-2.698	3.759	1,088

$$MDE_{80, 1SD}^{pp} = 100(t_{0.975, \nu} + \Phi^{-1}(0.8))SE(\hat{\beta})s_{\text{tone}}$$

Effect and minimum detectable effect (MDE) are return percentage points per one tone standard deviation. MDE approximates 80% power for a two-sided 5% test using the stated inference degrees of freedom. It measures precision rather than observed power.

All: a one-standard-deviation increase in proportional negative tone corresponds to -0.82 return percentage points ($p = 0.206$), with an approximate detectable effect of 1.84 points. The weighted-score effect is -1.06 points ($p = 0.223$).

10-K: a one-standard-deviation increase in proportional negative tone corresponds to -3.07 return percentage points ($p = 0.394$), with an approximate detectable effect of 10.31 points. The weighted-score effect is -3.11 points ($p = 0.296$).

10-Q: a one-standard-deviation increase in proportional negative tone corresponds to -2.59 return percentage points ($p = 0.043$), with an approximate detectable effect of 3.51 points. The weighted-score effect is -2.70 points ($p = 0.049$).

Statistical insignificance does not establish zero price response. Earnings releases can overlap the filing window and drive the same returns. Correlated specifications and multiple unadjusted tests also limit the interpretation of isolated significant estimates.

Report-type differences and limitations

The within-company trends supported by both weighting schemes at the 5% level are: 10-K negative tone increase. These receive more weight than results that depend on the weighting scheme, although the two measures are correlated. Volatility associations assess incremental information after existing risk is controlled; short-window returns have separate precision and earnings-overlap limitations.

Negative proportional-score SD is 0.421 percentage points for 10-Ks and 0.965 for 10-Qs. Report length, repeated language and changing content may affect dispersion; this analysis does not identify their separate contributions.

Uncertainty proportional-score SD is 0.239 percentage points for 10-Ks and 0.586 for 10-Qs. Report length, repeated language and changing content may affect dispersion; this analysis does not identify their separate contributions.

10-Ks provide annual business and risk disclosures, while 10-Qs provide quarterly updates. More words do not necessarily imply more new information. Different significance levels across forms do not establish that coefficients differ.

1 of 12 estimable within-company trend tests change 5% significance between company-only and two-way clustering. The two-way estimates use 20 time clusters across samples, so finite-sample inference remains approximate. All four tone trends, both volatility specifications and both return measures are reported for pooled and separate report-type samples.

Of 175 distinct securities held on May 6, 2021, 106 (60.6%) do not match the frozen 2026 holdings by normalized ticker or valid CUSIP. Identity changes, mergers and share classes mean these absences do not all represent company failures.

The 2026 holdings selection, mixed holdings dates and incomplete historical identifiers limit generalisation. Market-window and share-count availability impose additional selection on the outcome samples. Full-corpus word weights use later filings and the current dictionary is applied retrospectively. The estimates therefore describe retrospective conditional associations, not causal effects or an implementable live strategy.

Sources

Loughran and McDonald (2011), *Journal of Finance* 66(1), 35–65: When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks
Loughran–McDonald Master Dictionary, 1993–2025 release; updated March 2026; accessed September 9, 2026

SEC EDGAR: filings and company facts

Yahoo Finance: price, volume, corporate actions and market indices

ARK holdings archive

FRE-GY 7871 A course holdings and assignment materials

Illustrative SEC filing